

Introduction to Programming COMP101 2025-26

Lab-08

Objectives:

Exercise-01 Assignment-01 Modularised

OR

Work assignment-05

Exercise-01:

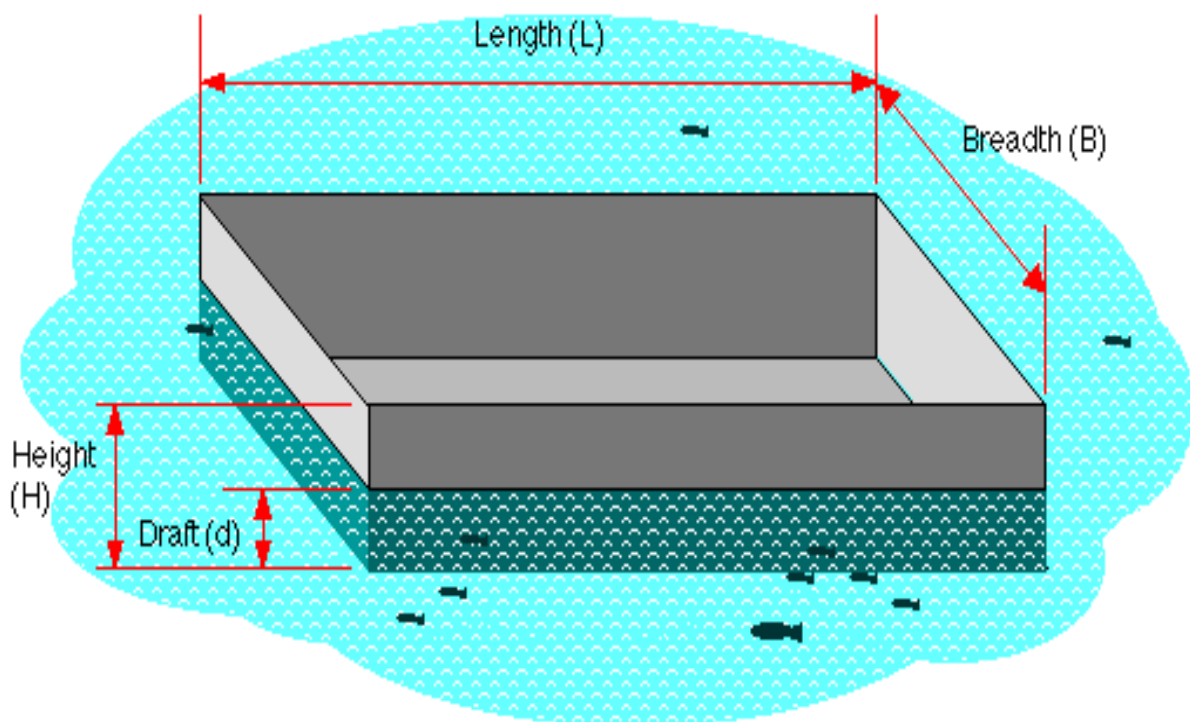
Modularise Assignment-01 (modified)

Develop and implement a Python program to accept three inputs to calculate the draft of a barge.

Given an iron barge defined in terms of inputs in metres for: length (L), breadth (B) and height (H), calculate and output the draft. (The draft is the depth by which the barge sinks into the water.

The barge is basically an 'empty box' with no lid. It has four walls and one floor.

External factors such as the superstructure, the density of the water, combined weight of personnel, cargo etc do not need to be factored in.



Input:

Three inputs for length, breadth and height with no validation

Process (Calculation):

a) iron weighs 1.06kg per metre squared

b) draft = mass of barge divided by: the length multiplied by the breadth

c) mass = surface area of barge multiplied by the weight of iron

d) surface area = total of area of each wall plus the area of the floor

Output:

The draft calculated

Exercise-01: SOLUTION

```
def data_in():
    length = int(input('Enter length: '))
    breadth = int(input('Enter breadth: '))
    height = int(input('Enter height: '))
    return length, breadth, height

def data_proc(length, breadth, height):
    iron = 1.06
    surface_area = ((length*height)*4) + (length*breadth)
    mass = surface_area * 1.06
    draft = mass/(length*breadth)
    return draft

def data_out(draft):
    print (draft)
    return ''

def main():
    length, breadth, height = data_in()
    calc = data_proc(length, breadth, height)
    data_out(calc)

main()
```

No user-oriented code used here to make the active code clear and readable for programmers.
User-oriented code can be added later

Exercise-01: SOLUTION ANALYSIS

In def main():

```
def main():
```

```
#call data_in to capture and return 3 values
```

```
#we need 3 vars to capture the return values
```

```
#There are no arguments in the call - we don't yet have any yet to pass to the function
```

```
length, breadth, height = data_in()
```

```
#call data_proc with 3 arguments and return 1 value
```

```
# we need 1 var to capture the 1 return value
```

```
calc = data_proc(length, breadth, height)
```

```
data_out(calc)
```

```
#Because data_out returns nothing, we don't need a variable to call it
```

```
#It passes one argument which is the draft calculated from data_proc
```

In def data_in():

```
#Called in def main() – we have to call it with three vars because it returns three values
```

```
#The call does not have any arguments – we don't have any to give it
```

```
#Because it is called without arguments, it does not have any parameters
```

```
Its job is to generate the values needed in other functions.
```

```
def data_in():
```

```
    length = int(input('Enter length: '))
```

```
    breadth = int(input('Enter breadth: '))
```

```
    height = int(input('Enter height: '))
```

```
    return length, breadth, height
```

In def data_proc():

#Called in def main() with three arguments length, breadth, height which were returned to the main body of code from data_in

#Because it is called with three arguments, it has three parameters

#Uses standard equations to calculate the draft to be used in data_out later

#It returns one value – hence we have to call it with one var in def main()

def data_proc(length, breadth, height):

 iron = 1.06

 surface_area = ((length*height)*4) + (length*breadth)

 mass = surface_area * 1.06

 draft = mass/(length*breadth)

 return draft

In def data_out():

#Called in def main() with one argument, calc which holds the value of the draft calculation returned to the main body of code from data_proc

#Because it is called with one argument, it has one parameter

#Simply prints out the parameter

#It returns no value so technically we don't need a variable to call it with – but keeps it consistent with the other functions

#Functions like this that do not return anything will return 'None' to the caller to show the function has ended without any return of values.

#The word 'None' can output on screen.

We can stop this output using a blank quote

def data_out(draft):

 print (draft)

 return ' '